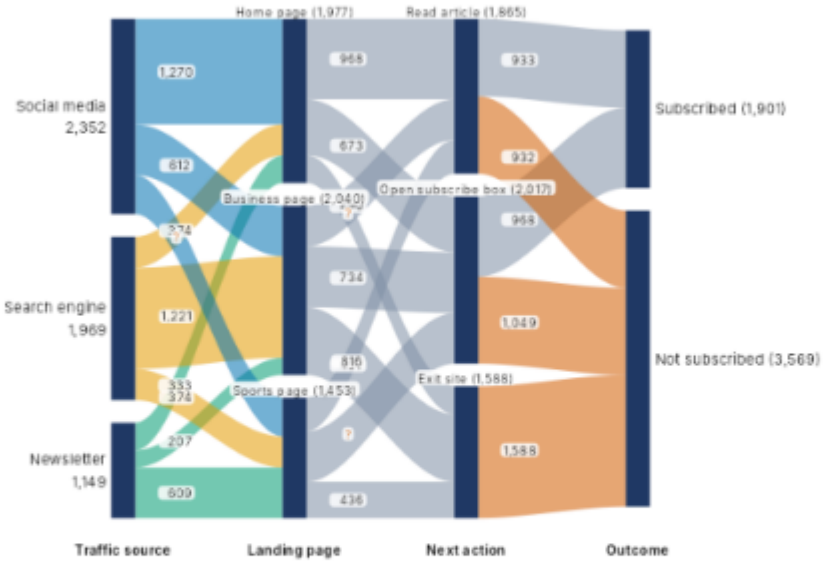
DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot D

R.No: 29 Reg. No: 192524136 Name: Peram Mahitha

Scenario: The analytics team of a news portal tracked 5,470 visits last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of visits on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Home page to Exit site; Sports page to Open subscribe box; Social media to Sports page. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Subscribed as a percentage of all visits; (ii) the largest single flow into "Not subscribed" or "Exit site"; (iii) the percentage of "Home page" visits that go on to "Read article". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 300 visits. Compute the height in cm of each landing-page node (Home page, Sports page, Business page). [2]

(e) Write R code: a links data frame for the first four flows leaving "Social media" and "Newsletter" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

(a) Three Hidden Flows

Using conservation of flow, the number of visitors entering a node must equal the number leaving it.

1. Home page to Exit site: 2,040 − 968 − 436 = 636 visitors
2. Social media to Open subscribe box: 612 − 274 = 338 visitors
3. Home page to Open subscribe box: 2,040 − 933 − 734 = 373 visitors

(b) Three Percentages

1. Social media to Sports page

$$\frac{374}{2352} \times 100 = 15.90\%$$

2. Open subscribe box to Subscribe

$$\frac{932}{2017} \times 100 = 46.21\%$$

3.Home page visits that go to Read article

$$\frac{968}{2040} \times 100 = 47.45\%$$

(c) Design Justification

- Colour rule: Different colours are used for traffic sources to distinguish Social media, Search engine and Newsletter visitors.
- Node height: The height of each node represents the number of visitors passing through that stage. Higher visitor counts are shown with taller nodes.
- Labels: Each node and flow is labelled with its name and visitor count, making it easier to understand the website traffic.
- What to highlight: The largest flow and the final outcomes, Subscribed and Not subscribed, should be highlighted to help the analytics team understand user behaviour.

(d) Landing Page Node Heights

Given: 1 cm of node height = 300 visits.

Formula:

$$\text{Height in cm} = \frac{\text{Number of visits}}{300}$$

1. Home page: 2040/300 =6.80 cm
2. Sports page: 1451/300 = 4.84 cm
3. Business page: 2040/300 = 6.80 cm

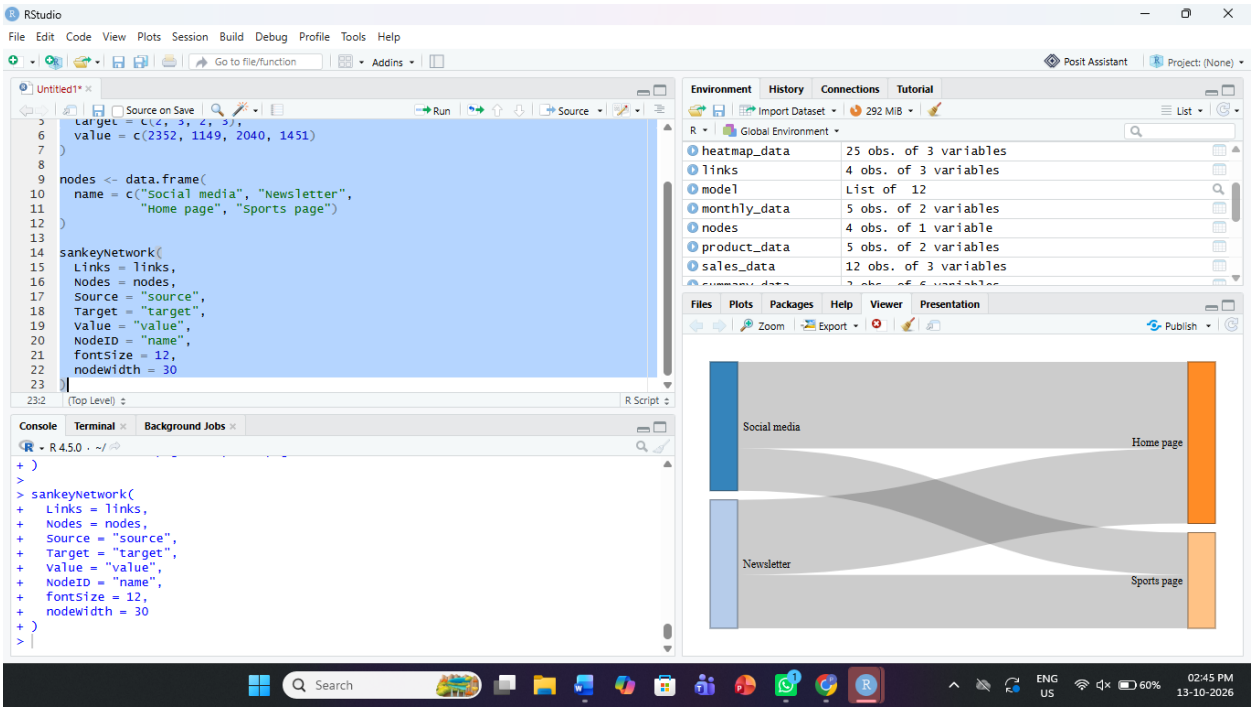
(e) R Code

```
library(networkD3)

links <- data.frame(
  source = c(0, 0, 1, 1),
  target = c(2, 3, 2, 3),
  value = c(2352, 1149, 2040, 1451)
)

nodes <- data.frame(
  name = c("Social media", "Newsletter",
           "Home page", "Sports page")
)

sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 12,
  nodeWidth = 30
)
```



(f) Two Situations Where Sankey Diagrams Mislead

1. Too many flows: When there are many overlapping flows, the diagram becomes crowded and difficult to read. An alternative is a grouped bar chart.

2. Very small flows: Small flows may become difficult to see when compared with large flows. An alternative is a table showing the exact visitor counts.